

0,1

Bits Manipulation

→ a = 42 ←

a =

int a = 42  
↓  
integer no.



bits  
↳ 0, 1

# Number System

↳  $(6784)_{(10)}$  base  $\rightarrow$  Decimal no. system.  
↳  $[0-9]$   
↳ 10 digits

$$\Rightarrow 6 \times 10^3 + 7 \times 10^2 + 8 \times 10^1 + 4 \times 10^0$$

$$\Rightarrow \begin{array}{r} 6000 \\ 700 \\ 80 \\ 4 \\ \hline 6784 \end{array}$$

# Binary Number System

$\hookrightarrow [0,1]$   $\Rightarrow$  (2) digits

$\hookrightarrow$  base = 2

Computer

$(101101)_2$

bns

$$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$



↳ Binary → Decimal  
→ Decimal → Binary

$$(45)_{10} \rightarrow (?)_2$$

$$(95)_{10} \rightarrow (?)_2$$

↳  $(\boxed{101101})_2 \rightarrow (?)_{10}$

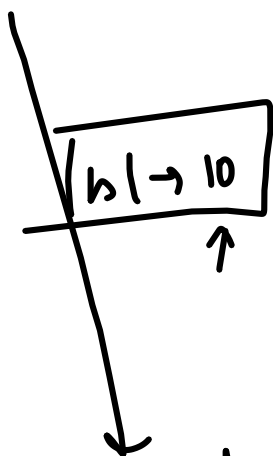
*32 bit*

↳  $(101001)_2 \rightarrow (?)_{10}$

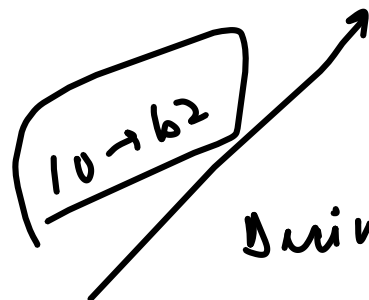
Q.

Base ( $b_1$ )  $\longrightarrow$  Base ( $b_2$ )

any base to  
decimal



decimal



Decimal to any base.

Addition of 2 decimal nos.

|           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|
| $10/10=1$ | $13/10$   | $10/10$   | $12/10$   |           |
| $10/10=2$ | 1         | 1         | 1         |           |
| 0         | 6         | 4         | 5         | 9         |
| 0         | 3         | 8         | 4         | 3         |
| $1^0/10$  | $10^0/10$ | $13^0/10$ | $10^0/10$ | $12^0/10$ |
| 1         | 0         | 3         | 0         | 2         |

$$9+3=12$$

$$12^0/10 = 2 \quad \text{base}$$

$$12/10 = 1$$

(base = 2)

Additional of 2 binary nos.

|          |         |         |         |          |         |         |
|----------|---------|---------|---------|----------|---------|---------|
| $11/2=0$ | $3/2=1$ | $2/2=1$ | $2/2$   | $11/2=0$ | $1/2=0$ |         |
|          |         |         | 1       |          |         |         |
| 0        | 1       | 1       | 0       | 1        | 0       | 1       |
| 0        | 1       | 1       | 1       | 1        | 1       | 0       |
| $1^0/2$  | $3^0/2$ | $3^0/2$ | $2^0/2$ | $2^0/2$  | $1^0/2$ | $1^0/2$ |
| 1        | 1       | 1       | 0       | 0        | 1       | 1       |

$$1 \text{ int} = 4 \text{ bytes}$$

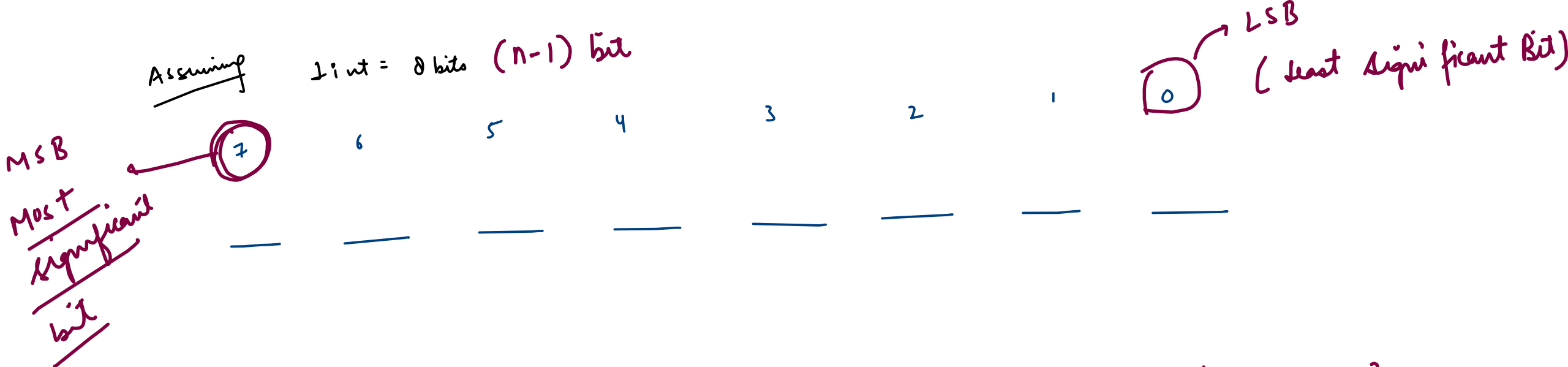
$$1 \text{ byte} = 8 \text{ bits}$$

$1 \text{ int} = 4 \times 8 = 32 \text{ bits}$

 $\Rightarrow ?$

↳ In order to represent any integer value we need 32 bits

— — — — — ' — / — ' — ' — 0 — / — / — ' — 0 — 1 — 0 —



$n=10$

??  
 $n = -10$

$\hookrightarrow \text{MSB} = 1$   
 $\hookrightarrow$  2's complement  
 $\hookrightarrow$  1's complement

$n = -10$

⑦

6

5

4

3

2

1

0

1

0

0

0

1

0

1

0

↓

$$1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + \underline{1} \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$\Rightarrow 128 + 0 + 0 + 0 + 8 + 0 + 2 + 0$$

$$\Rightarrow 128 + 8 + 2 \Rightarrow 138 \neq \textcircled{-10}$$

$\Rightarrow$  whenever we want to represent a -ve no.  $\Rightarrow$

2's complement

MSB = 1

Obs:-

sign bit way of represent a negative no. is a wrong concept.

↳ 2's complement

$$\boxed{1's(n) + 1}$$

↳ flip of bit  $1 \leftrightarrow 0$

$$\boxed{A = -10}$$

|         |    |   |   |   |   |   |   |   |   |
|---------|----|---|---|---|---|---|---|---|---|
| ↳       | 10 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|         | 0  | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
|         | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
|         | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1's(10) | 1  | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
|         | 1  | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| +       | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

---

|  |   |   |   |   |   |   |   |   |
|--|---|---|---|---|---|---|---|---|
|  | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|  | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 |

2's(10)  
↓  
-10

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

↓

$$\cancel{128} + 64 + 32 + 16 + 0 + 4 + 2 + 0 \neq -10$$

$$-128 + 118 \Rightarrow \boxed{-10}$$

↓  
**MSB is always -ve**

Obs:-

$$\begin{array}{r} 64 \\ 32 \\ \hline 96 \\ 16 \\ \hline 112 \\ + 4 \\ \hline 116 \\ + 2 \\ \hline 118 \end{array}$$

Range

$$-128 + \underline{64 + 32 + 16 + 8 + 4 + 2 + 1} \Rightarrow 127 \Rightarrow -128 + 127 = \boxed{-1}$$

8 bits

| 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>1</u> | <u>1</u> | <u>1</u> | <u>1</u> | <u>1</u> | <u>1</u> | <u>1</u> |

Max  $\rightarrow$

$$0 \times (-2^7) \Rightarrow 0 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 \Rightarrow \boxed{127}$$

| 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> |

Min  $\rightarrow$

$$-128 + 0 + 0 + 0 + 0 + 0 + 0 + 0 = \boxed{-128}$$

Range of num. for 8 bits  $\Rightarrow \boxed{[-128, 127]}$



$$\boxed{[-2^7, 2^7 - 1]} \Rightarrow 2$$



1 int = 4 bytes = 4 × 8 = 32 bits.

↳  $[-2^{31}, 2^{31}-1]$

64 bits

↳  $[-2^{63}, 2^{63}-1]$

12 bits

↳  $[-2^{11}, 2^{11}-1]$

In general

n bits

↳  $[-2^{n-1}, 2^{n-1}-1]$

int a =

long b =

$$15 \approx 10^8$$

15.

$$\frac{2^{10}}{10^3} = 1024 \approx 1000$$

$$10^3 = 1000$$

$$2^{10} \approx 10^3$$

Take cube on both sides.

$$2^{30} \approx 10^9$$

$$2^{31} - 1 \approx 10^9$$

integer  
range information

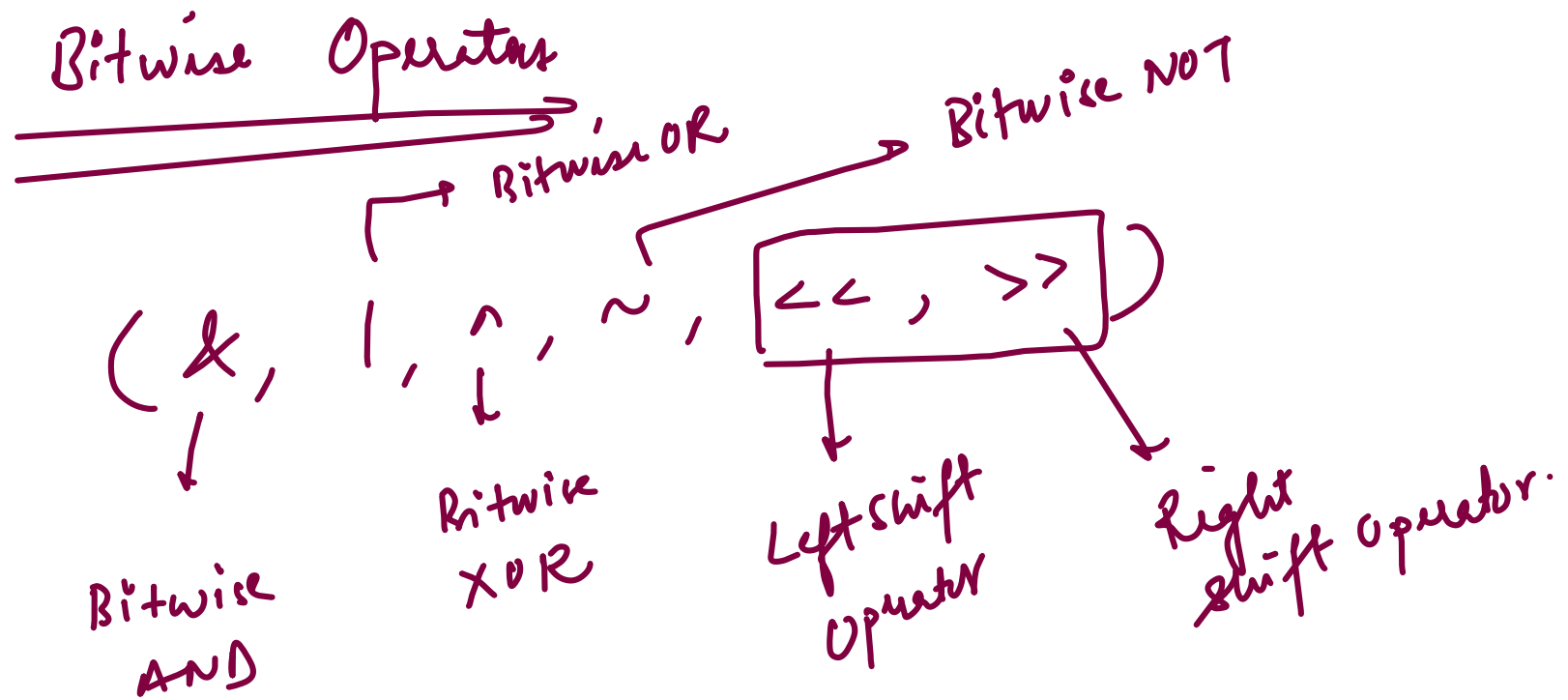
curve forints

$$10^9 \leq \text{num} \leq 10^9 - 1$$

~~0(n)~~  $\rightarrow$   $\overline{TLÉ}$

$$O(n)$$

## ↳ Bitwise Operators



|  | a | b | $a \& b$ | $a   b$ | $a \wedge b$ | $\sim a$ | $\sim b$ |
|--|---|---|----------|---------|--------------|----------|----------|
|  | 0 | 0 | 0        | 0       | 0            | 1        | 1        |
|  | 0 | 1 | 0        | 1       | 1            | 1        | 0        |
|  | 1 | 0 | 0        | 1       | 1            | 0        | 1        |
|  | 1 | 1 | 1        | 1       | 0            | 0        | 0        |

→ 5 min. 410.26 pm

$\sim a \Rightarrow 15$

$2^{15} \Rightarrow \boxed{\sim a + 1}$

ON  $\rightarrow 1$ , OFF  $\rightarrow (0)$

↳ &

↳ when all the bits are ON (1)

↳  $0 | P \Rightarrow \underline{1}$

↳ 0

↳ 1 when any bit is ON (1)

↳  $0 | P \Rightarrow \underline{1}$

↳ 0

↳ ^

↳ when both the bits are

same

↳  $0 | P \Rightarrow 0$

↳ 1

$$a = 24$$

$$b = 18$$

$$a \oplus b \Rightarrow ?$$

$$a \mid b \Rightarrow ?$$

$$a \wedge b$$

$$\sim a$$

$$8 \text{ min}$$

|  | 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |                      |
|--|----------|----------|----------|----------|----------|----------|----------|----------|----------------------|
|  | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u> |                      |
|  | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> |                      |
|  | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | $\Rightarrow 16$     |
|  | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> | $\Rightarrow 26$     |
|  | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> | $\Rightarrow 10$     |
|  | <u>1</u> | <u>1</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>1</u> | <u>1</u> | $\Rightarrow \equiv$ |

$$64 - 128 + 64 + 32 + 0 + 0 + 4 + 2 + 1 =$$

## Bit manipulation properties

$$\textcircled{1} \left\{ \begin{array}{l} a \& b = b \& a \\ a | b = b | a \\ a \wedge b = b \wedge a \end{array} \right.$$

$\textcircled{2}$

$$\begin{aligned} a \& b \& c &= a \& c \& b = b \& a \& c \\ a | b | c &= a | c | b = b | a | c \\ a \wedge b \wedge c &= a \wedge c \wedge b = b \wedge a \wedge c \end{aligned}$$

$a = 10$      $(\text{print}(a \& 1))$

1    0    1    0

0    0    0    1

---

0    0    0    0

---

$\Rightarrow$  even  $\Rightarrow$  0    (even & 1)

$\Rightarrow$  odd  $\Rightarrow$  1    (odd & 1)

$\rightarrow$

$a = 11$      $\text{print}(a \& 1)$

1    0    1    1

0    0    0    1

---

0    0    0    1

$\text{if}( \underline{a \& 1} == 0 ) \rightarrow a \text{ is an even no.}$

$\text{if}( a \& 1 == 1 ) \rightarrow a \text{ is an odd no.}$

$$\Rightarrow a \& a \Rightarrow \underline{a}$$

$$\Rightarrow a \wedge \underline{a} = \underline{0}$$

$$\Rightarrow a | a = \underline{a}$$

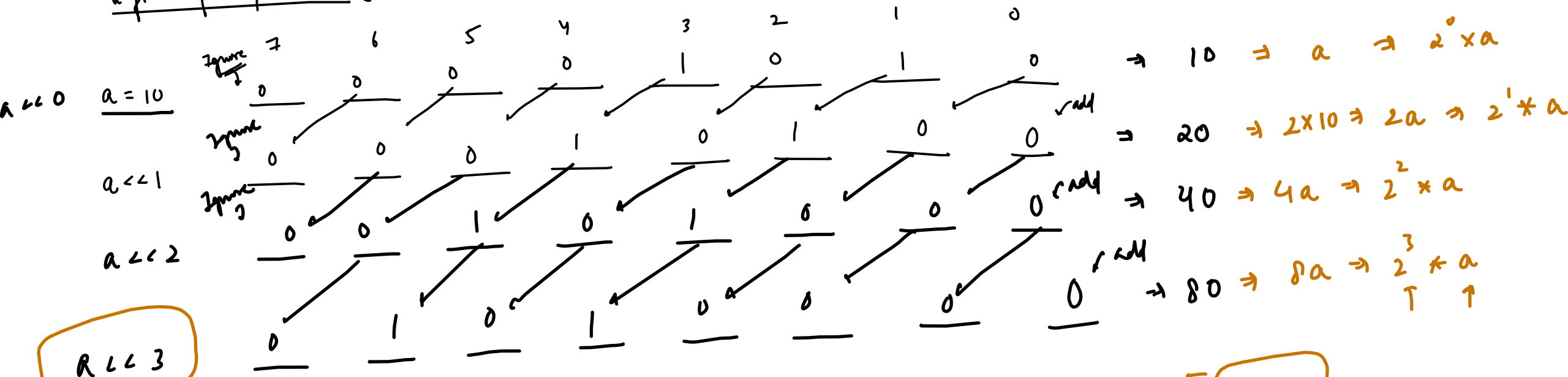
$$\Rightarrow a \& 0 = \underline{0}$$

$$\Rightarrow a | 0 = \underline{a} //$$

$$a = \begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ \hline 1 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}$$



# Left shift Operator ( $\ll$ )



$$1 \ll 5 \Rightarrow 2^5 \Rightarrow 32$$

~~Imp:~~

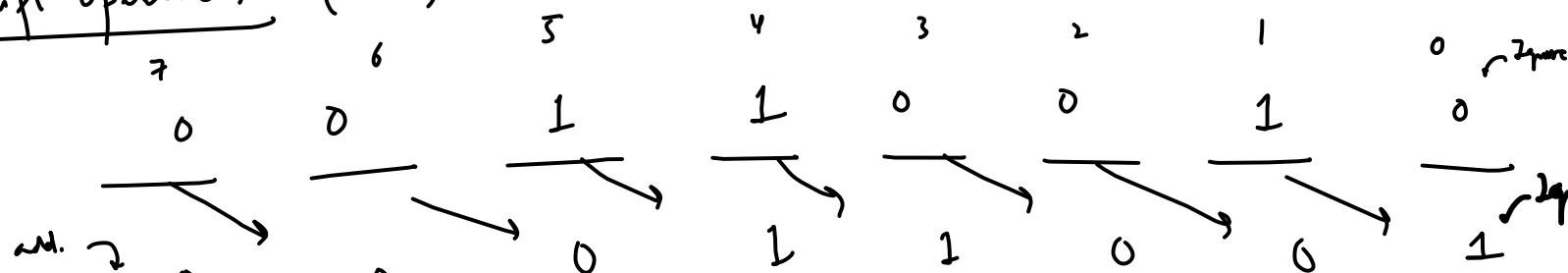
$$a \ll n \Rightarrow 2^n \times a$$

if  $a = 1$

$$1 \ll n \Rightarrow 2^n$$

Right shift Operator :- ( $\gg$ )

$a \gg 0$   
 $a = 50$



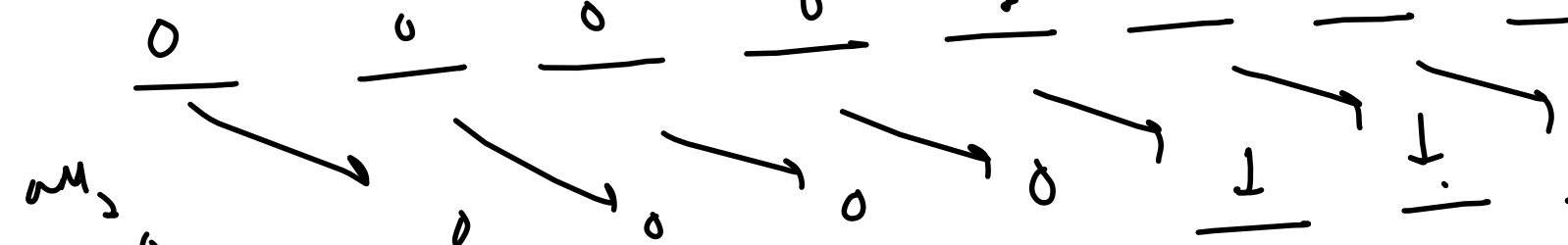
$\Rightarrow 50 \Rightarrow a \Rightarrow a/2^0$

$a \gg 1$



$\Rightarrow 25 \Rightarrow a/2 \Rightarrow a/2^1$

$a \gg 2$



$\Rightarrow 12 \Rightarrow a/4 \Rightarrow a/2^2$

$a \gg 3$



$\Rightarrow 6 \Rightarrow a/8 \Rightarrow a/2^3$

$$a \gg n \Rightarrow a/2^n$$

$$\text{if } a = 1 \Rightarrow 1/2^n$$

Q- Given N and i, check the i<sup>th</sup> bit is set or not.

$\frac{\text{set}}{\downarrow}$   
 1

$\frac{\text{not}}{\downarrow}$   
 0

⇒ set ⇒ true

⇒ not ⇒ false.

LSO | RSO

N = 10

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

↑ set.  
 ↑ not

i = 2 ⇒ false.

i = 3 ⇒ true

$N \& (1 \ll i) > 0 \Rightarrow \text{set}$

using left shift operator ( $\ll$ )

N = 10

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

1 < 3 ↑

mask. ⇒

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |

[2<sup>3</sup>]

N & mask ⇒

70

4 set

$N=10$

$i=2$

mask

| 2        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |
| <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>0</u> |

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> |
|----------|----------|----------|----------|----------|----------|----------|----------|

$$N \& (1 \ll i) == 0$$

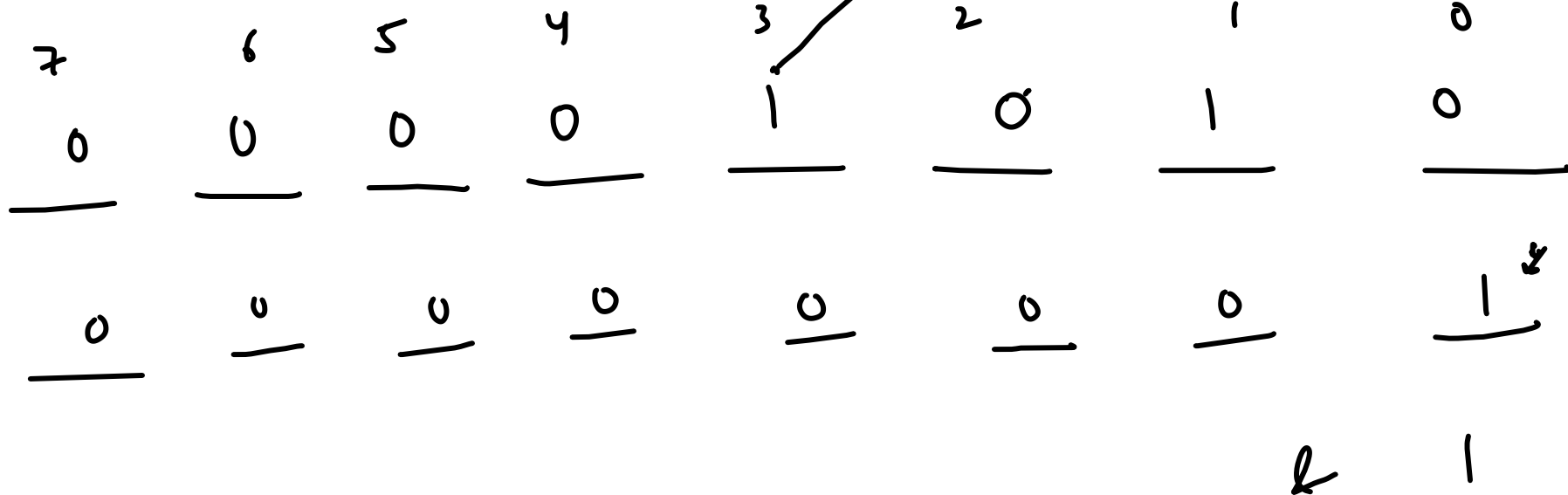
$\Rightarrow$

not set  
=

Using Right Shift Operator ( $\gg$ )

$N=10$

$i=3$



$N \gg i$

$(N \gg i) \& 1 == 1$

$\hookrightarrow$  set

$\frac{1}{0} \Rightarrow$  set.  
 $\frac{0}{0} \Rightarrow$  unset.

⇒ To check the  $i$ th bit is set or not

5 min

using left shift operator



$$N \& (1 \ll i) > 0 \Rightarrow \text{set.}$$

$$N = 45$$

$$\underline{i = 4}$$

$$\underline{i = 6}$$

using right shift operator



$$((N \gg i) \& 1) == 1 \rightarrow \text{set}$$

⇒ set the i<sup>th</sup> bit

$N = 10$

| 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |

$i = 5$

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> |
|----------|----------|----------|----------|----------|----------|----------|----------|

$N / \text{mask} \Rightarrow$

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |
|----------|----------|----------|----------|----------|----------|----------|----------|

↑  
set

$N / (1 \ll i)$

⇒ Clear the i<sup>th</sup> bit

→  $N = 10$   
 $i = 3$

mask

→  $\sim \text{mask}$

$N \& (\sim \text{mask})$

|       | 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|
|       | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |
|       | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u> |
|       | <u>1</u> | <u>1</u> | <u>1</u> | <u>1</u> | <u>0</u> | <u>1</u> | <u>1</u> | <u>1</u> |
| <hr/> |          |          |          |          |          |          |          |          |
|       | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> | <u>0</u> |

$N \& (\sim (1 \ll i))$

≠



Toggle the  $i^{\text{th}}$  bit

$N = 10$

$i = 4$  mask

| 7         | 6         | 5         | 4         | 3         | 2         | 1         | 0        |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|
| <u>0.</u> | <u>0</u>  | <u>0.</u> | <u>0.</u> | <u>1.</u> | <u>0.</u> | <u>1.</u> | <u>0</u> |
| <u>0.</u> | <u>0.</u> | <u>0.</u> | <u>1.</u> | <u>0.</u> | <u>0.</u> | <u>0.</u> | <u>0</u> |
| <hr/>     |           |           |           |           |           |           |          |
| <u>0</u>  | <u>0</u>  | <u>0</u>  | <u>1.</u> | <u>1.</u> | <u>0</u>  | <u>1</u>  | <u>0</u> |

$N \wedge (1 \ll i)$



Single w.

2, 2, 1

$$\boxed{2^2} \Rightarrow 0^1 = \underline{\underline{1}}$$

$3 \% 3 = 0$

$[10, 10, 8, 8, 8, 7, 10]$   
 $\rightarrow$  return 7

→ 10  
10  
8  
8  
8  
7  
10

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 5 | 0 | 0 | 1 | 0 | 1 | 0 |
| 4 | 0 | 0 | 1 | 0 | 1 | 0 |
| 3 | 0 | 0 | 1 | 0 | 1 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 |

$1+1+0+0+0+1+1$   
 $\rightarrow 1$

$10^n, 10^n, 8^n, 8^n, 8^n, 7^n, 10^n$   
 $\rightarrow ?$

freq array  
 $4 \times 10 - 3$   
 $4 \times 8 - 3$   
 $4 \times 7 - 1$   
 $\rightarrow 10 \% 3 = 1 \times 0$   
 $8 \% 3 = 0$

$4 \% 3 = 1$   
 $1 \% 3 = 0$   
 $4, 1, 6, 1, 1, 2$

0 0 0 1 1 1

$32 \times O(n) \rightarrow O(32n)$   
 $\rightarrow O(n)$   
 $4 + 2 + 1 = 7$   
 $S.C. \rightarrow O(1)$

$$i = 3$$

$sum = 1 \quad 1 \cdot 3 = 1$

$$sum < i$$

$$1 < 3 \Rightarrow 2^3 \Rightarrow 8$$

$$2$$

$$a < 2 \Rightarrow a \times 2^2 \Rightarrow$$

$$\Rightarrow 2 \times 2^2 \Rightarrow 2^3 = 8$$

$ans = ans + (sum < i)$

ans =

|   |                                                                                                                                             |               |               |               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|
|   |                                                                                                                                             |               |               | 0             |
|   |                                                                                                                                             |               | 1             |               |
|   |                                                                                                                                             | 2             |               |               |
|   | 3                                                                                                                                           |               |               |               |
|   | $\frac{0}{1}$                                                                                                                               | $\frac{0}{1}$ | $\frac{0}{1}$ | $\frac{1}{0}$ |
|   |                                                                                                                                             |               |               |               |
| 1 | <div style="display: flex; justify-content: space-around; width: 100%;"> <span>1</span> <span>0</span> <span>0</span> <span>1</span> </div> |               |               |               |

Single no. - 3

→ 2 times  
→ 2 dec → ① time

$[1, 2, 1, 3, 2, 5]$   
↑ ↑ ↑ ↑ ↑ ↑

3, 5  
5, 3

↳  $1 \wedge 2 \wedge 1 \wedge 3 \wedge 2 \wedge 5 \Rightarrow 3 \wedge 5$   
↓  
xor → 1 time

⇒

|       |          |          |          |          |
|-------|----------|----------|----------|----------|
|       | 3        | 2        | 1        | 0        |
| 3 →   | <u>0</u> | <u>0</u> | <u>①</u> | <u>1</u> |
| 5 →   | <u>0</u> | <u>1</u> | <u>0</u> | <u>1</u> |
| <hr/> |          |          |          |          |
|       |          |          | <u>①</u> |          |
|       | <u>0</u> | <u>1</u> | <u>①</u> | <u>0</u> |
| <hr/> |          |          |          |          |
|       |          |          | ↑        |          |

|     |          |          |          |
|-----|----------|----------|----------|
|     | 2        | 1        | 0        |
| 1 → | <u>0</u> | <u>0</u> | <u>1</u> |
| 2 → | <u>0</u> | <u>1</u> | <u>0</u> |
| 3 → | <u>0</u> | <u>1</u> | <u>1</u> |
| 5 → | <u>1</u> | <u>0</u> | <u>1</u> |

③      ⑤

$2 \wedge 3 \wedge 2$

bit is set

$1 \wedge 1 \wedge 5$

bit is unset.

101

[ 1, 2, 5, 7, 8, 1, 2, 5, 7, 9 ]

XOR  $\rightarrow$   $1^{\wedge} 2^{\wedge} 5^{\wedge} 7^{\wedge} 8^{\wedge} 1^{\wedge} 2^{\wedge} 5^{\wedge} 7^{\wedge} 9 \Rightarrow 1111$  X

$1 \rightarrow 01$   
 $2 \rightarrow 10$

$\Rightarrow 8^{\wedge} 9$

$1^{\wedge} 5^{\wedge} 7^{\wedge} 9$   
 $1$

| 3        | 2        | 1        | 0 ✓        |
|----------|----------|----------|------------|
| <u>1</u> | <u>0</u> | <u>0</u> | <u>0</u>   |
| <u>1</u> | <u>0</u> | <u>0</u> | <u>1</u>   |
| <hr/>    |          |          |            |
| <u>0</u> | <u>0</u> | <u>0</u> | <u>1</u> ← |

right most set bit.

$\Rightarrow$  To find out the right most set bit

$\Rightarrow$   $x \& (-x)$

Input: arr = [1,3,4,8],  
 queries = [[0,1],[1,2],[0,3],[3,3]]

Output: [2,7,14,8]

3^4

for (query) {

→ T.C →  $O(Q \times N)$

→ T.C →  $O(Q + N)$

prefixXOR[3]

↳ 1^3^4^8

prefixXOR[3] ^ prefixXOR[2]

→ 1^3^4^8 ^ 1^3^4

→ 8

↳ prefix and suffix

↓ suffix sum

( prefix sum

prefix multiply

prefix XOR

Prefix XOR

| 0 | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 3 | 4 | 8 |

| 0     | 1     | 2     | 3       |
|-------|-------|-------|---------|
| 1     | 1^3   | 1^3^4 | 1^3^4^8 |
| (0-0) | (0-1) | (0-2) | (0-3)   |

start end  
 (1-2)

[0,1] ⇒

[0,1] ⇒

prefixXOR[2] ^ prefixXOR[1-1] → O(1)

1^3^4 ^ 1 ⇒ 3^4

prefixXOR[1] ^ prefixXOR[0-1]

X

invalid

prefixXOR[1] = 1^3

| 0 | 1   | 2 | 3 |
|---|-----|---|---|
| 1 | 3   | 4 | 8 |
| 1 | 1^3 |   |   |

prefixXOR[2] = prefixXOR[1] ^ arr[2]

1^3^4

Start = 0 , n = 5

$$0 + 2 \times 1 = \underline{2}$$

$$2 + 2 \times 1 = 4$$



# Kernighan's Algorithm

Count of set bits:

count = 1



count = 1  
N = 10  $\Rightarrow$   
N-1 = 9  $\Rightarrow$

|   |   |   |   |     |   |   |   |
|---|---|---|---|-----|---|---|---|
| 7 | 6 | 5 | 4 | 3   | 2 | 1 | 0 |
| ↓ | ↓ | ↓ | ↓ | ↓ ✓ |   | ✓ |   |
| 0 | 0 | 0 | 0 | 1   | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1   | 0 | 0 | 1 |

count = 2 N = 8  
N-1 = 7 //

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

N = 0

while (N != 0) {

count++;

N = (N & N-1);

}

# Total Hamming Distance

$$\text{arr} = [4, 14, 2]$$

$$\begin{array}{l} 4 \Rightarrow \begin{array}{cccc} \underline{0} & \underline{1} & \underline{0} & \underline{0} \end{array} \\ 14 \Rightarrow \begin{array}{cccc} \underline{1} & \underline{1} & \underline{1} & \underline{0} \end{array} \\ 2 \Rightarrow \begin{array}{cccc} \underline{0} & \underline{0} & \underline{1} & \underline{0} \end{array} \end{array}$$

$$2 + 2 + 2 = \textcircled{6}$$

$$\begin{array}{r} 4 \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \\ 14 \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{0} \\ \hline \quad \quad 1 \quad \quad 0 \quad 1 \quad 0 \rightarrow \textcircled{2} \\ \hline \end{array}$$

$$\begin{array}{r} 4 \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \rightarrow \textcircled{2} \\ 2 \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad 0 \\ \hline \quad \quad 0 \quad \quad 1 \quad 1 \quad 0 \\ \hline \end{array}$$

$$\begin{array}{r} 14 \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{0} \rightarrow \textcircled{2} \\ 2 \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad 0 \\ \hline \quad \quad 1 \quad \quad 1 \quad 0 \quad 0 \\ \hline \end{array}$$

$n = \text{arr.length}$

|                  |              |              |              |              |
|------------------|--------------|--------------|--------------|--------------|
|                  | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |
|                  | 3            | 2            | 1            | 0            |
| 4 $\Rightarrow$  | <u>0</u>     | <u>1</u>     | <u>0</u>     | <u>0</u>     |
| 14 $\Rightarrow$ | <u>1</u>     | <u>1</u>     | <u>1</u>     | <u>0</u>     |
| 2 $\Rightarrow$  | <u>0</u>     | <u>0</u>     | <u>1</u>     | <u>0</u>     |

|       |   |
|-------|---|
| $i=0$ | 0 |
| $i=1$ | 2 |
| $i=2$ | 2 |
| $i=3$ | 1 |

| $\text{CSB}$ | $\text{CSB} (n - \text{CSB})$ |
|--------------|-------------------------------|
| 3            | $\Rightarrow 0 \times 3 = 0$  |
| 1            | $\Rightarrow 2 \times 1 = 2$  |
| 1            | $\Rightarrow 2 \times 1 = 2$  |
| 2            | $\Rightarrow 1 \times 2 = 2$  |
|              | <u><u>6</u></u>               |

$O(n^2)$   
 $\downarrow$   
 $O(32 \times n)$   
 $\approx \underline{\underline{O(n)}}$

$\downarrow$   
 $\text{for (int } i=0; i < 32; i++) \{$   
 $\quad \text{for (int } i=0; i < n; i++) \{$

# Power of Two

$$\boxed{n \& (n-1) == 0} \rightarrow \text{power of 2}$$

$$n = 8$$

$$\begin{array}{cccc} \underline{1} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$\begin{array}{cccc} \underline{0} & \underline{1} & \underline{1} & \underline{1} \end{array}$$

$$n-1=7$$

$$\begin{array}{cccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$n = 16$$

$$\begin{array}{ccccc} 4 & 3 & 2 & 1 & 0 \\ \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$\begin{array}{ccccc} \underline{0} & \underline{1} & \underline{1} & \underline{1} & \underline{1} \end{array}$$

$$n-1=15$$

$$\begin{array}{ccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$h = 4$$

$$\begin{array}{cc} 3 & 2 \\ \underline{0} & \underline{1} \end{array} \quad \begin{array}{cc} 1 & 0 \\ \underline{1} & \underline{0} \end{array}$$

$$n-1=5$$

$$\begin{array}{cc} \underline{0} & \underline{0} \\ \underline{1} & \underline{1} \end{array}$$

$$\begin{array}{cc} \underline{0} & \underline{0} \\ \underline{1} & \underline{0} \end{array}$$

$$\boxed{h = 16}$$

$$h-1=7$$

$$\begin{array}{cccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} \\ \underline{1} & \underline{1} & \underline{1} & \underline{1} \end{array}$$

Power of four  $\rightarrow$  Power of Two +  $\text{num} \% 3 == 1$

4  $\rightarrow$  16  $\rightarrow$   $n=8$  X

$4^1$   $\rightarrow$   $4^2$

$\downarrow$   $\downarrow$

$(2^2)^1$   $(2^2)^2$

$\rightarrow 2^2$   $2^4$   $n=2$

$$4 \div 3 = 1$$

$$16 \% 3 = 1$$

$$64 \% 3 = 1$$

$$128'_{10} = 1$$

$n=10$

$\downarrow$   $\leftarrow$   
1 0 1 0

$n=11$

1 0 1 1

1

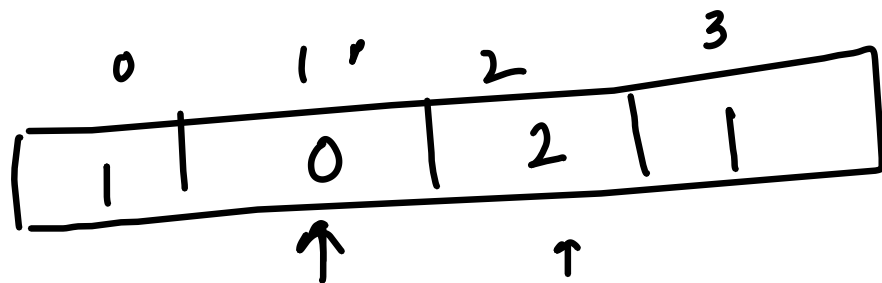
---

1 0 1 1

encoded array  $\rightarrow [1, 2, 3]$   
                                   $\uparrow \uparrow$

first = 1

$1^{\wedge} 1$


$$\begin{array}{r} 10 \\ 11 \\ \hline 01 \end{array}$$

Missing no.

$n = 3$

$[0 - 3]$

4 0, 1, 2, 3

[3, 0, 1]

~~3~~ ^ ~~1~~ ^ ~~1~~ ^ ~~0~~ ^ ~~1~~ ^ 2 ^ ~~3~~

⇒ 8

0 1 2 3 4 5 6 7 8  
[9, 6, 4, 2, 3, 5, 7, 0, 1]

~~0~~ ^ ~~1~~ ^ ~~2~~ ^ ~~3~~ ^ ~~4~~ ^ ~~5~~ ^ ~~6~~ ^ 7 ^ ~~8~~ ^ ~~9~~ ^ ~~0~~ ^ ~~1~~ ^ ~~2~~ ^ ~~3~~ ^ ~~4~~ ^ ~~5~~ ^ ~~6~~ ^ ~~7~~ ^ ~~8~~ ^ ~~9~~



Start = 10

, goal = 7

|          |          |          |          |
|----------|----------|----------|----------|
| <u>0</u> | <u>1</u> | <u>1</u> | <u>0</u> |
| <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |

⇒ (3)

|          |          |          |          |
|----------|----------|----------|----------|
| <u>0</u> | <u>1</u> | <u>1</u> | <u>1</u> |
|----------|----------|----------|----------|

|          |          |          |          |
|----------|----------|----------|----------|
| <u>1</u> | <u>0</u> | <u>1</u> | <u>0</u> |
| <u>0</u> | <u>1</u> | <u>1</u> | <u>1</u> |

---

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 0 | 1 |
|---|---|---|---|

---

→ unset bit

$\Rightarrow$  perm  $[1, 3, 2]$   $\leftarrow n$  , first  $N$  num.

included  $[1^3, 3^2]$

$(2, 1)$

$1 - N$   $\downarrow$   
 $1 \rightarrow$  first ele of perm array.

$[2, 1]$

$[1, 3, 2]$

$1^2$

$\begin{array}{r} 01 \\ 10 \\ \hline 11 \end{array} \rightarrow 3$

$\begin{array}{r} 01 \\ 11 \\ \hline 10 \end{array}$

$\begin{array}{r} 11 \\ 10 \\ \hline 01 \end{array}$

firstde

$1-N$

$p \rightarrow [1, \underline{3}, \underline{2}, \underline{3}, 4]$

$e \rightarrow [2, 1, 1, 7] [n]$

$[a, b, c, d, e] [n+1]$

$$a^b = 2$$

$$b^c = 1$$

$$c^d = 1$$

$$d^e = 7$$

$$\begin{array}{r} 10 \\ 11 \\ \hline 000 \\ 011 \\ \hline 1421 \end{array}$$

$$\text{first} = 1^2^3^4^5^2^1$$

$$\text{first} = (a^b c^d e)^{(a^b c^d)}$$



first  $\rightarrow (a^n b^n c) \wedge (a^n b)$

$(a^n b^n c) \wedge (b^n c)$

encoded =  $[2, 1]$

$n = 2$

$n+1 = 3$

$a^n b = 2$

$b^n c = 1$

$1^2 = 3$

first =  $(\cancel{1}^n 2^n 3) \wedge (\cancel{1})$

$perm = [i, 3, 2]$

$$\begin{array}{r} 10 \\ 11 \\ \hline 01 \end{array}$$

⇒

arr = [2, 1]

perm ⇒ [1, 3, 2]

(1-N)

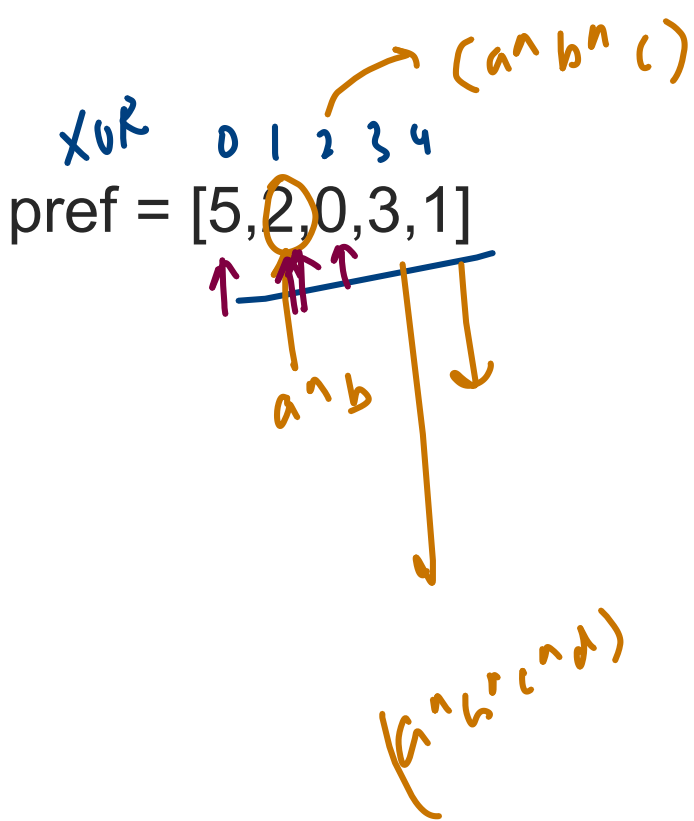
$1^3 = 2 \rightarrow a^b$   
 $3^2 = 1 \rightarrow b^c$

first ⇒ 1

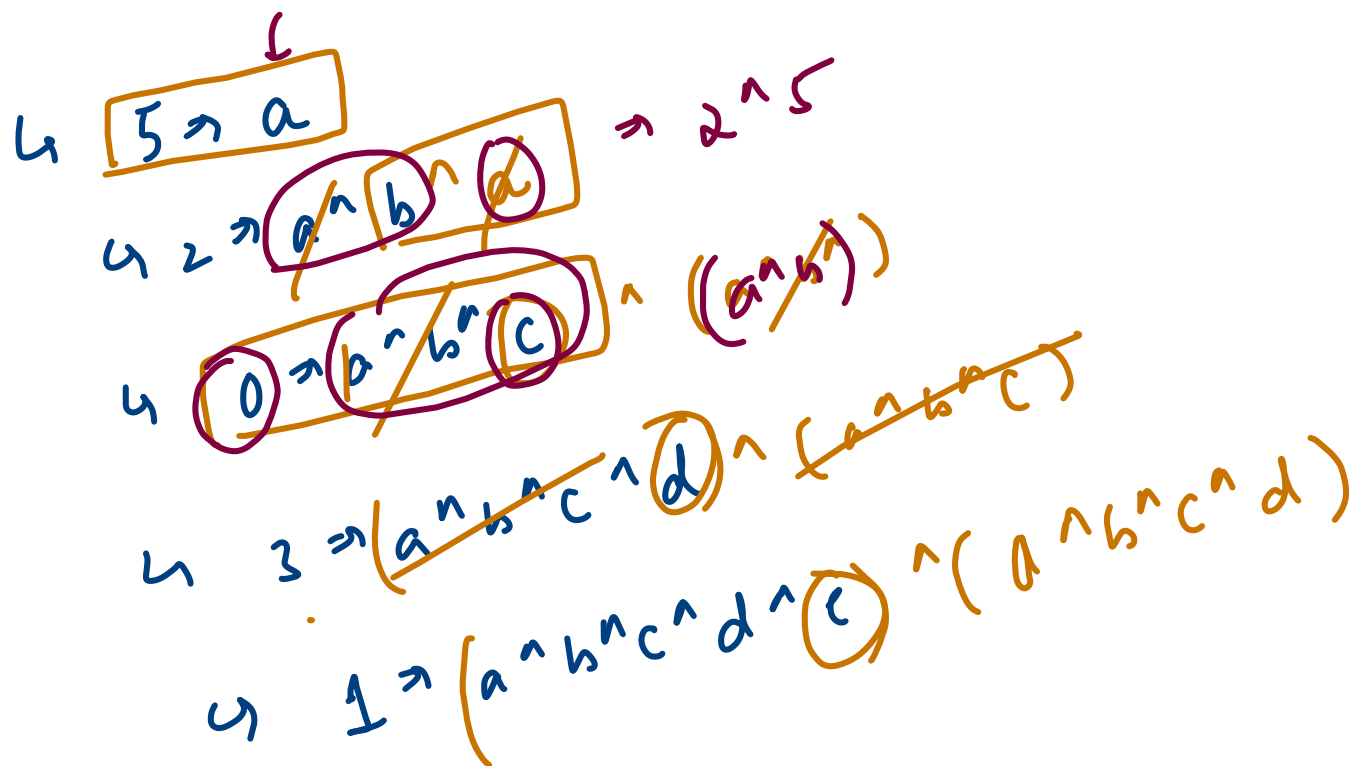
$$1^2 = 3$$

$$3^1 = 2$$

first ⇒  $(a^b)^c \wedge (b^c)^a \rightarrow a$



arr = [a, b, c, d, e]



arr

|   |       |       |   |   |
|---|-------|-------|---|---|
| 0 | 1     | 2     | 3 | 4 |
| 5 | $2^5$ | $2^0$ |   |   |

arr[i] = prefix[i]  $\wedge$  prefix[i-1]

XOR

a

↑

↓

nums = [2,1,3,4], k = 1

$$\text{C++}(\text{XOR}^k) \rightarrow$$

$$1 \wedge 1 = 0 \wedge 0$$

$$2 \rightarrow 0 \ 1 \ 0 \ /$$

$$1 \rightarrow 0 \ 0 \ 1 \ /$$

$$3 \rightarrow 0 \ 1 \ 1 \ /$$

$$4 \rightarrow 1 \ 0 \ 0 \ /$$

$$0 \ 1 \ 0 \ 1 \ /$$

$$0 \ 0 \ 1$$

$$1 \wedge 1 \wedge 0$$

$$\Rightarrow \underline{\underline{2}}$$

XOR

$$1 \ 0 \ 0 \ /$$

K

$$0 \ 0 \ 1 \ /$$

$$\underline{\underline{1 \ 0 \ 1}}$$

$$\downarrow$$

$$\underline{\underline{\text{LSB}}}$$

ans = 0

→ true | false

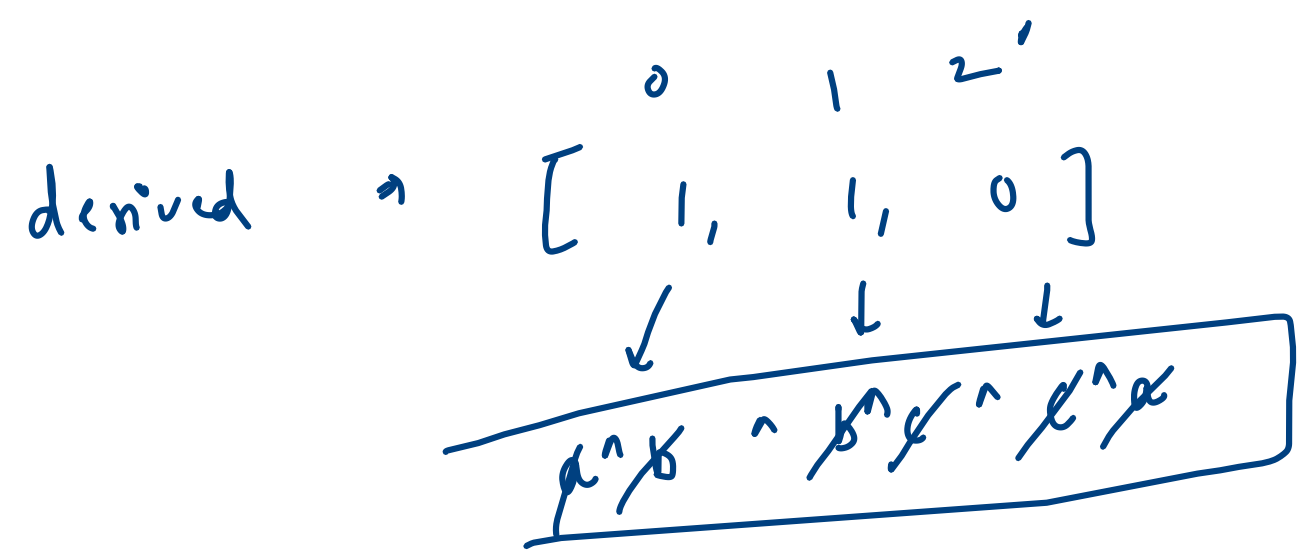
derived = [1, 1, 0]

$$\text{derived}[i] = \text{original}[i] \wedge \text{original}[i+1]$$

if  $i = n-1$   
 $\text{derived}[i] = \text{original}[i] \wedge \text{original}[0]$

original

original → <sup>0</sup>a, <sup>1</sup>b, <sup>2</sup>c





ans [1-N]  
n=11

solve(n)

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 | ✓ |
| 2 | 0 | 0 | 1 | 0 | ✓ |
| 3 | 0 | 0 | 1 | 1 | ✓ |
| 4 | 0 | 1 | 0 | 0 | ✓ |
| 5 | 0 | 1 | 0 | 1 | ✓ |
| 6 | 0 | 1 | 1 | 0 | ✓ |
| 7 | 0 | 1 | 1 | 1 | ✓ |

- 8
- 9
- 10
- 11

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |

$\Rightarrow$  ans power of 2  $\leq 11$   
 $x=3$   
 $2^3 = 8$   
 $4 \times \frac{8}{2} + \frac{8}{2} + \frac{8}{2} + 0 =$   
 $\Rightarrow 4+4+4 = 12 + 4 =$

$n - 2^3 + 1$   
 $11 - 8 + 1 \Rightarrow 3 + 1 = 4$   
 $n - 2^3$   
 $11 - 8 = 3$  solve

$\Rightarrow$  closest power of 2  $\leq n$   
 $2^x$   
 $\Rightarrow$  (  $2^{3^r} \Rightarrow x=3$  )

$\Rightarrow x * (2^{x-1}) \Rightarrow$  ans.

$\Rightarrow n - 2^x + 1 \Rightarrow$

$\Rightarrow [x * 2^{x-1}] + [n - 2^x + 1] + \text{solve}(n - 2^x)$

$12 + 4 + 1 + 2 + 1 \rightarrow 20$

$2^1$   
 $1 * 2^{1-1} \Rightarrow 1 * 1 \Rightarrow 1$   
 $3 - 2^1 + 1 \Rightarrow 3 - 2 + 1 \Rightarrow 2$   
 $\text{solve}(3 - 2^1) = 1$

$11 - 8 + 1 = 4$   
 ① if(n==0)  

|   |   |   |   |
|---|---|---|---|
| 1 | 0 | 0 | 1 |
| 2 | 0 | 1 | 0 |
| 3 | 0 | 1 | 1 |

$n = 17$

|      | 4 | 3 | 2 | 1 | 0 |
|------|---|---|---|---|---|
| 1 →  | 0 | 0 | 0 | 0 | 1 |
| 2 →  | 0 | 0 | 0 | 1 | 0 |
| 3 →  | 0 | 0 | 0 | 1 | 1 |
| 4 →  | 0 | 0 | 1 | 0 | 0 |
| 5 →  | 0 | 0 | 1 | 0 | 1 |
| 6 →  | 0 | 0 | 1 | 1 | 0 |
| 7 →  | 0 | 0 | 1 | 1 | 1 |
| 8 →  | 0 | 1 | 0 | 0 | 0 |
| 9 →  | 0 | 1 | 0 | 0 | 1 |
| 10 → | 0 | 1 | 0 | 1 | 0 |
| 11 → | 0 | 1 | 0 | 1 | 1 |
| 12 → | 0 | 1 | 1 | 0 | 0 |
| 13 → | 0 | 1 | 1 | 0 | 1 |
| 14 → | 0 | 1 | 1 | 1 | 0 |
| 15 → | 0 | 1 | 1 | 1 | 1 |
| 16 → | 1 | 0 | 0 | 0 | 0 |
| 17 → | 1 | 0 | 0 | 0 | 1 |

$$2^4 \rightarrow 16 < 17$$

$$\Rightarrow x = 4$$

$$2^x (2^{n-x} - 1) \Rightarrow 4 * (2^3) \Rightarrow 4 * 8 = 32$$

$$\Rightarrow n - 2^x + 1 = 17 - 2^4 + 1 \Rightarrow 17 - 16 + 1 = 2$$

$$\Rightarrow \text{solve}(n - 2^x) \Rightarrow \text{solve}(17 - 16) \Rightarrow \text{solve}(1)$$

↓  
1

solve(1)

$$2^0 \rightarrow 1$$

$$\downarrow$$
  

$$\text{solve}(n - 2^0) \Rightarrow \text{solve}(0)$$

↓

0

$$\Rightarrow 32 + 2 + 1 \Rightarrow$$

35

[1,2,3], k = 2

↳ ①

→ shortest subarray.

→ bitwise OR  $\geq k$

~~1  $\geq 2$~~

1 → 0 1  
2 → 1 0

3 → 1 1  
3 → 1 1

③

[1,2,3] =

3  $\geq k$

1 → 0 1  
3 → 1 1  
1 1 → 3  $\geq 2$

1 → 0 1  
2 → 1 0  
1 1 → ③

3  $\geq 2$

⇒ ② → [1,2]

2 → 1 0  
3 → 1 1  
1 1 → 3  $\geq 2$

2 →  $\geq 2$

3 →  $\geq 2$

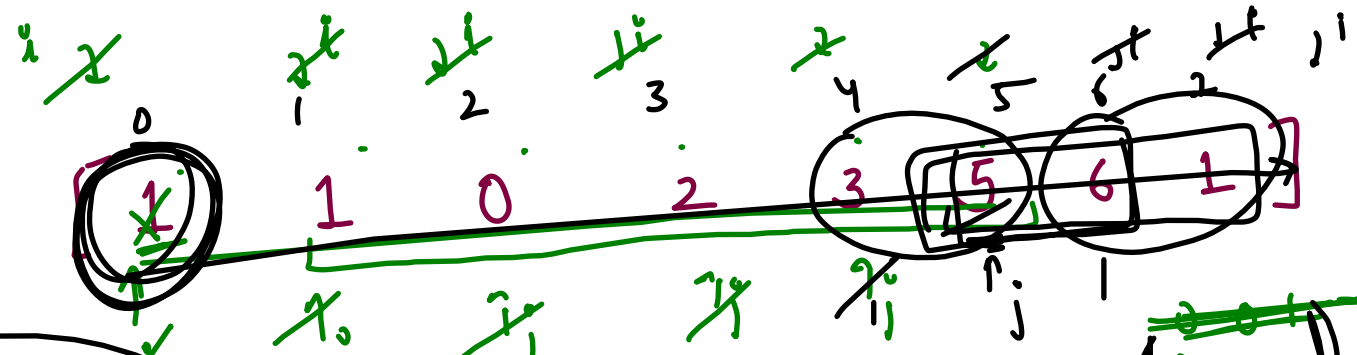
⇒ [1] ①  
①

Generate all the  
subarray  $\geq k$   $O(n^3)$

$k = 7$

← [ 1 1 0 2 3 5 6 1 ]

~~$10^5 \times 10^5 + 10^5 \Rightarrow 10^{15} > 10^8$~~



$k = 7$

sliding window

$O(n)$

$minLen = 2$

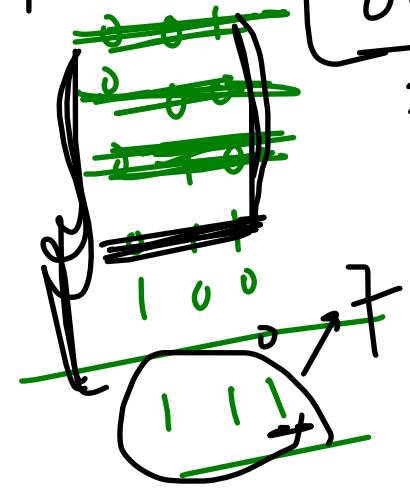
$arr[i]$

$ans = 0$

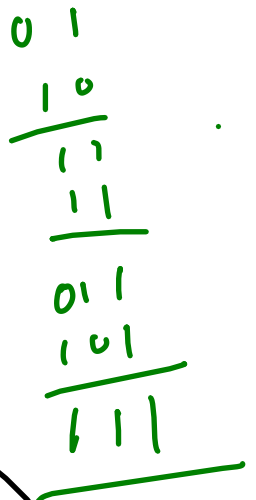
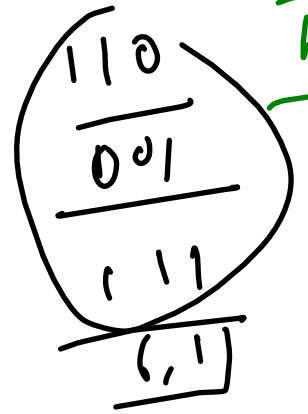
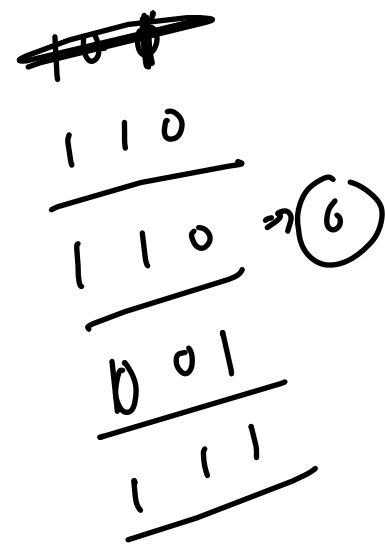
- $i = 0 \Rightarrow arr[0] = 1$
- $i = 1 \Rightarrow arr[1] = 1$
- $i = 2 \Rightarrow arr[2] = 0$
- $i = 3 \Rightarrow arr[3] = 2$
- $i = 4 \Rightarrow arr[4] = 3$
- $i = 5 \Rightarrow arr[5] = 5$

- $0 | 1 = 1$
- $1 | 1 = 1$
- $1 | 0 = 1$
- $1 | 2 = 3$
- $3 | 3 = 3$
- $3 | 5 = 7$

- 3, 5
- 5, 6
- 6, 1



$s \geq k$



$$a = 7$$

$$b = 2$$

$$a + b = 9$$

$$a = \begin{array}{ccc} 2 & 1 & 0 \\ \hline 1 & 1 & 1 \end{array}$$

$$b = \begin{array}{ccc} & & \\ \hline 0 & 1 & 0 \end{array}$$

$$a \ll n \rightarrow a \times 2^n$$

$$2 \ll 1 \rightarrow 2 \times 2^1 = 4$$

$$a = 1$$

$$4 \times 2^1 \rightarrow 8$$

$$\begin{array}{ccc} 1 & 0 & 1 \\ \hline 1 & 0 & 0 \end{array}$$

$$a \rightarrow \begin{array}{ccc} 0 & 0 & 1 \end{array}$$

$$\text{carry} \rightarrow \begin{array}{ccc} 1 & 0 & 1 \\ \hline 1 & 0 & 0 \end{array}$$

$$b \rightarrow \begin{array}{ccc} 1 & 0 & 0 \end{array}$$

$$b = 4 \ll 1$$

$$a = 1$$

$$b = 8$$

$$a \rightarrow \begin{array}{ccc} 1 & 0 & 0 & 1 \end{array}$$

$$\text{carry} \rightarrow \begin{array}{ccc} 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 \end{array}$$

$$\text{while}(b \neq 0)$$

$$0 \ll 1 \rightarrow 0$$

$a \wedge b \Rightarrow$  sum of  $a$  and  $b$  but without carry.

carry  $\Rightarrow a \& b$

carry  $\Rightarrow a \& b$

$$\begin{array}{ccc} 1 & 1 & 1 \\ \hline 0 & 1 & 0 \end{array}$$

$$\rightarrow \begin{array}{ccc} 0 & 1 & 0 \end{array}$$

$$b \ll 1 \rightarrow$$

→

while ( b != 0 ) {

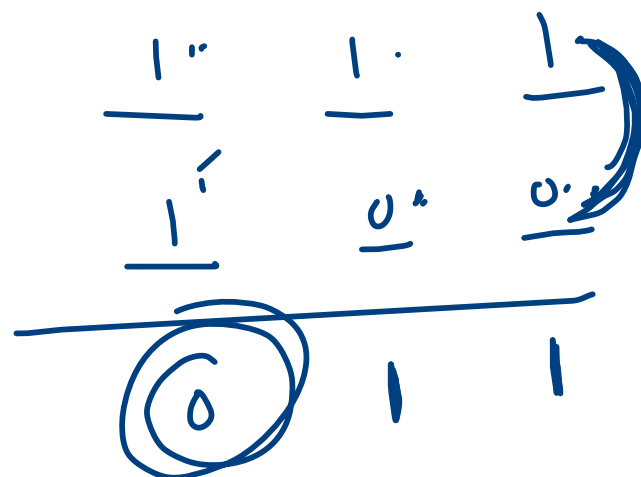
carry = a & b;

    a = a ^ b;

b = carry << 1

}

return a;



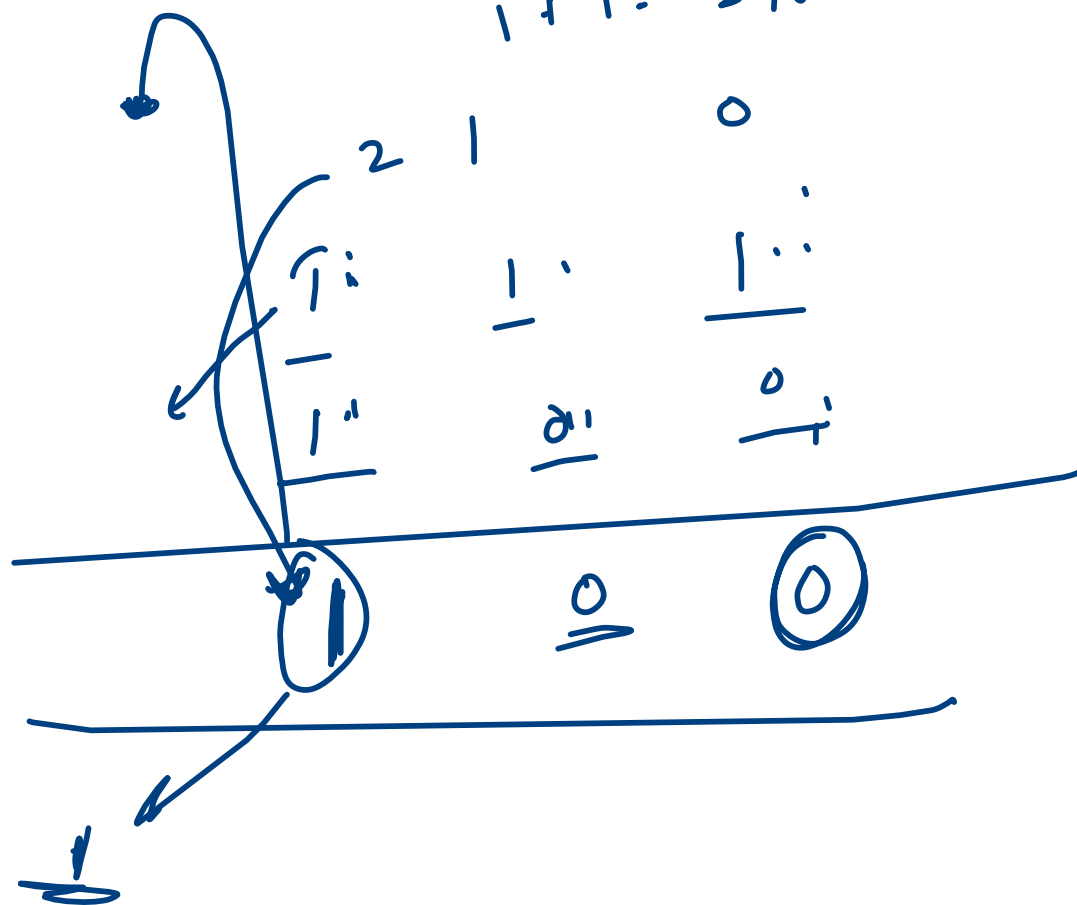
$$0 + 1 = 0$$

$$1 + 0 = 0$$

$$1 + 1 = 0$$

$$0 + 0 = 0$$

$$1 + 1 = 2 \text{ / } 2 = 0$$



# VIF-8 validation

↳ 1 byte → 0 x x x x x x x

↳ 2 bytes → 1 0 0 x x x x

1 0 x x x x x x

↳ 3 bytes → 1 1 0 0 x x x x

↳ 4 bytes → 1 1 1 0 x x x x 1 0 x x x x x x 1 0 x x x x x x



2 bits.

|   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|--|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |  |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |  |

$\frac{1}{-} \quad \frac{0}{-}$

11110 --

10



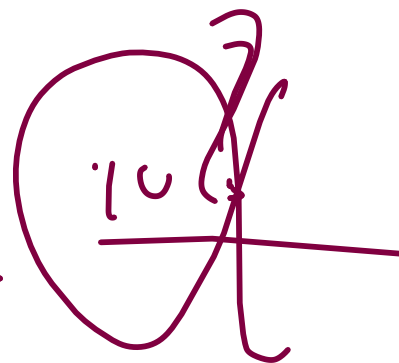
4 bits.

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

→ true.

1110

10



3 bits.

No=1

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |

X

```

class Solution {
    public String addBinary(String a, String b) {
        String ans = "";
        int carry = 0, i = a.length()-1, j = b.length()-1;
        while(i >= 0 || j >= 0 || carry > 0) {
            int sum = (i >= 0 ? a.charAt(i) - '0' : 0) + (j >= 0 ? b.charAt(j) - '0' : 0) + carry;
            int rem = sum % 2;
            ans = (rem + "") + ans;
            carry = sum / 2;
            i--;
            j--;
        }
        return ans;
    }
}

```

$a = \begin{matrix} & 0 & 1 & 2 & 3 \\ & 1 & 0 & 1 & 0 \\ \uparrow & \uparrow & \uparrow & \uparrow \\ i & & & \end{matrix}$ 
 $b = \begin{matrix} & 0 & 1 & 2 & 3 \\ & 1 & 0 & 1 & 1 \\ & & & \uparrow & \\ & & & j & \end{matrix}$

Carry = 0,  $i = 3$ ,  $j = 3$

$ans = \text{10101}$

$sum = 0 + 1 + 0 = 1$      $rem = 1 \% 2 = 1$      $carry = 1 / 2 = 0$

$sum = 1 + 1 + 0 = 2$      $rem = 2 \% 2 = 0$ ,  $carry = 2 / 2 = 1$

$sum = 0 + 0 + 1 = 1$      $rem = 1 \% 2 = 1$      $carry = 1 / 2 = 0$

$sum = 1 + 1 + 0 = 2$      $rem = 2 \% 2 = 0$      $carry = 2 / 2 = 1$

$sum = 0 + 0 + 1 = 1$      $rem = 1 \% 2 = 1$      $carry = 1 / 2 = 0$

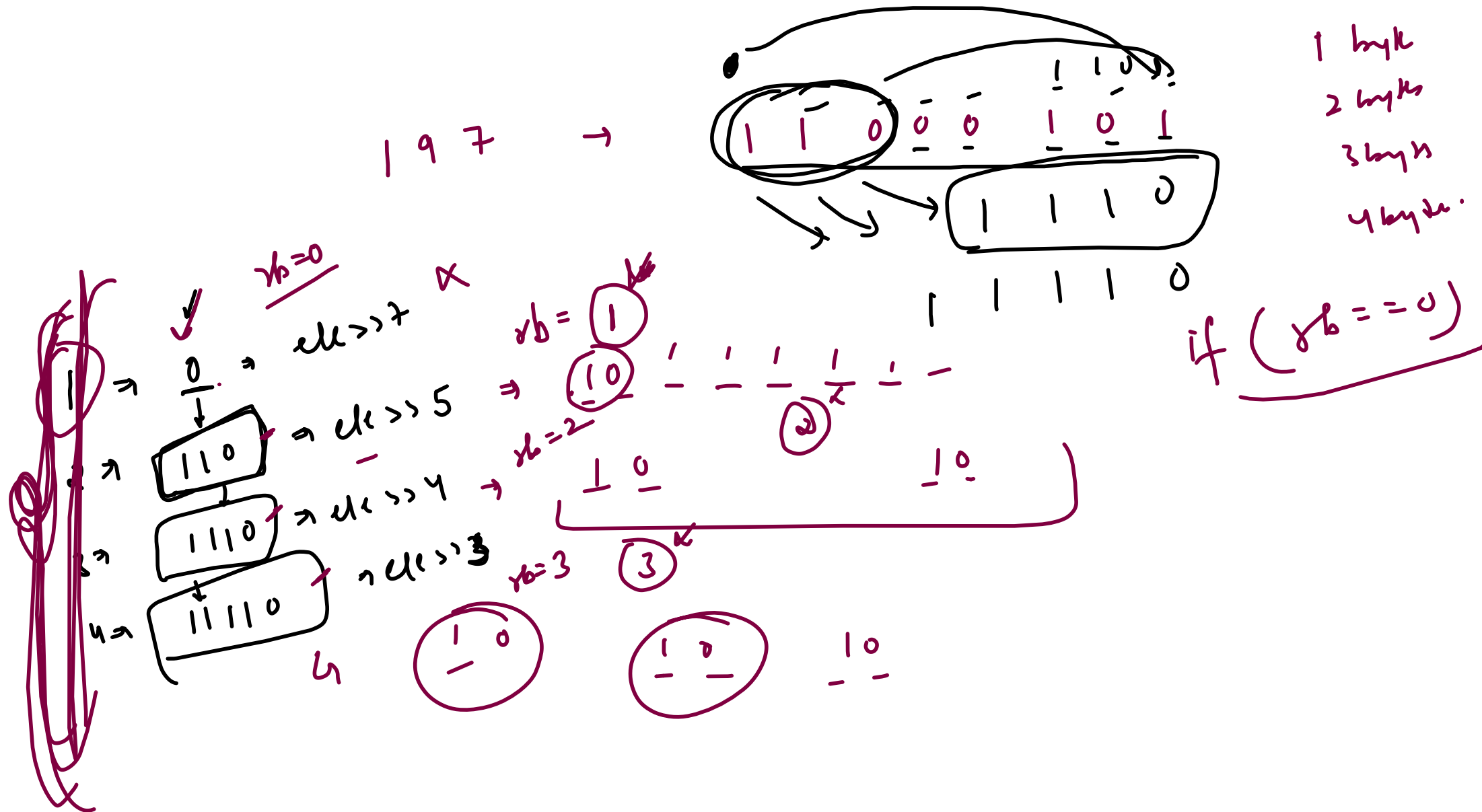
Input: a = "1010", b = "1011"  
 Output: "10101"

Input: data = [197,130,1]

Output: true

Explanation: data represents the octet sequence: 11000101 10000010 00000001.

It is a valid utf-8 encoding for a 2-bytes character followed by a 1-byte character.

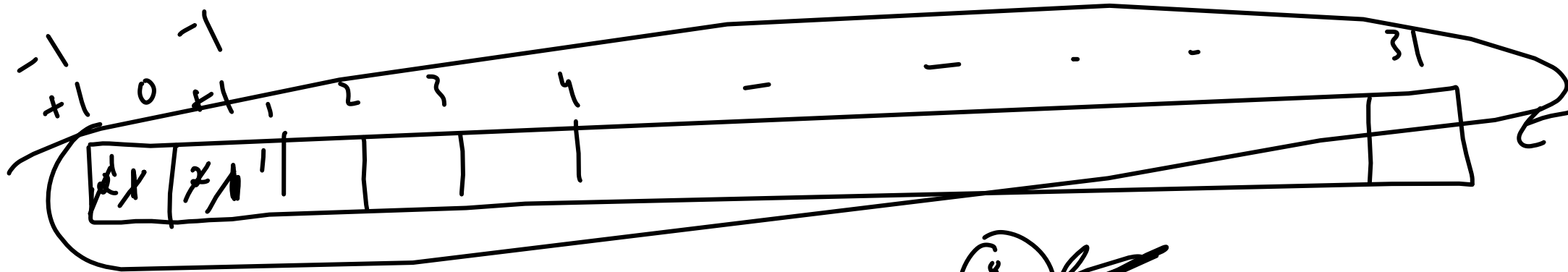


lint  $\rightarrow$  32 bits.

$\begin{matrix} \uparrow & \downarrow & \uparrow \\ 1 & 2 & 3 \end{matrix}$

$k=2$

$640 \geq k$



acquire  $\rightarrow$  (4)  
releasing  $\rightarrow$  (1)

0 1  
1 0  
1 1